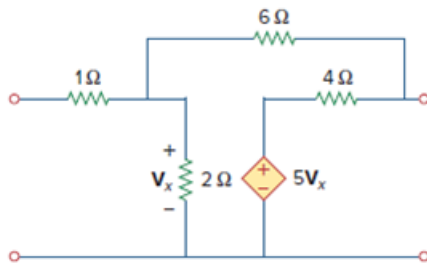


Problem

Obtain the **ABCD** parameters for the network in Fig. 19.102.

Figure 19.102



Step-by-step solution

Step 1 of 5

Consider the following network:

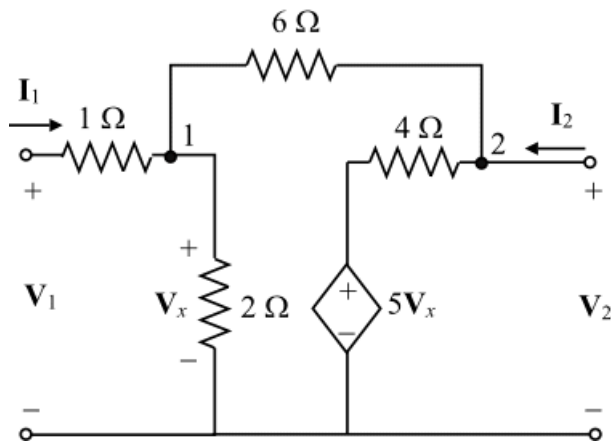


Figure 1

Step 2 of 5

Define ABCD parameters as follows.

$$V_1 = AV_2 - BI_2$$

$$I_1 = CV_2 - DI_2$$

Thus,

$$A = \left. \frac{V_1}{V_2} \right|_{I_2=0}$$

$$B = - \left. \frac{V_1}{I_2} \right|_{V_2=0}$$

$$C = \left. \frac{I_1}{V_2} \right|_{I_2=0}$$

$$D = - \left. \frac{I_1}{I_2} \right|_{V_2=0}$$

Step 3 of 5

To determine **A** and **C** parameters, apply voltage source V_1 and leave the output port open.

Apply Kirchhoff's current law at node 1.

$$\frac{V_1 - V_x}{1} = \frac{V_x}{2} + \frac{V_x - 5V_x}{6 + 4}$$

$$\frac{V_1 - V_x}{1} = \frac{V_x}{2} - \frac{4V_x}{10}$$

$$V_1 = 0.5V_x - 0.4V_x + V_x$$

$$V_1 = 1.1V_x$$

$$V_x = \frac{1}{1.1} V_1$$

$$= 0.909V_1$$

Apply Kirchhoff's voltage law to the output loop.

$$V_2 = 4\left(\frac{V_x - 5V_x}{6 + 4}\right) + 5V_x$$

$$= 4(-0.4V_x) + 5V_x$$

$$= -1.6V_x + 5V_x$$

$$= 3.4V_x$$

Substitute $0.909V_1$ for V_x .

$$V_2 = (3.4)(0.909V_1)$$

$$= 3.0906V_1$$

$$\frac{V_1}{V_2} = \frac{1}{3.0906}$$

$$= 0.3235$$

$$\text{Thus, } A = \left. \frac{V_1}{V_2} \right|_{I_2=0} = 0.3235$$

Step 4 of 5

Also,

$$I_1 = \frac{V_1 - V_x}{1}$$

And $V_2 = 3.4V_x$,

Substitute $\frac{V_2}{3.4}$ for V_x in the expression for I_1 .

$$I_1 = V_1 - \frac{V_2}{3.4}$$

Substitute $\frac{V_2}{3.0906}$ for V_1 obtained from the relation in step 2.

$$\begin{aligned} I_1 &= \frac{V_2}{3.0906} - \frac{V_2}{3.4} \\ &= 0.02944V_2 \end{aligned}$$

Thus, $C = \left. \frac{I_1}{V_2} \right|_{I_2=0} = 0.02944$

To determine **B** and **D** parameters, apply voltage source V_1 and leave the output port shorted.

Apply Kirchhoff's current law at node 1.

$$\frac{V_1 - V_x}{1} = \frac{V_x}{6} + \frac{V_x}{2}$$

$$V_1 = V_x + \frac{V_x}{6} + \frac{V_x}{2}$$

$$V_1 = \frac{10}{6}V_x$$

$$V_x = \frac{6}{10}V_1$$

Apply Kirchhoff's voltage law to the input loop.

$$V_1 = (1)I_1 + V_x$$

$$V_1 = I_1 + V_x$$

Substitute $\frac{10}{6}V_x$ for V_1 .

$$\frac{10}{6}V_x = I_1 + V_x$$

$$\frac{4}{6}V_x = I_1$$

$$V_x = \frac{6}{4}I_1$$

Also, as the output port is short circuited, write the expression for output current by applying Kirchhoff's current law at node 2.

$$\begin{aligned} I_2 &= -\frac{5V_x}{4} - \frac{V_x}{6} \\ &= -\frac{17}{12}V_x \end{aligned}$$

Substitute $\frac{6}{4}I_1$ for V_x .

$$I_2 = \left(-\frac{17}{12}\right)\left(\frac{6}{4}I_1\right)$$

$$= -\frac{17}{8}I_1$$

$$I_1 = -0.4706I_2$$

Thus, $D = -\left. \frac{I_1}{I_2} \right|_{V_2=0} = 0.4706$

Use the following relations.

$$V_1 = \frac{10}{6} V_x$$

$$V_x = \frac{6}{4} I_1$$

Thus,

$$\begin{aligned} V_1 &= \left(\frac{10}{6} \right) \left(\frac{6}{4} I_1 \right) \\ &= \frac{10}{4} I_1 \end{aligned}$$

Also,

$$I_2 = -\frac{17}{8} I_1$$

Determine **B**.

$$\begin{aligned} B &= -\frac{V_1}{I_2} \bigg|_{V_2=0} \\ &= -\frac{\left(\frac{10}{4} I_1 \right)}{-\frac{17}{8} I_1} \\ &= \left(\frac{10}{4} \right) \left(\frac{8}{17} \right) \\ &= 1.176 \end{aligned}$$

Thus, the **ABCD** parameters of the network are $\begin{bmatrix} 0.3235 & 1.176 \, \Omega \\ 0.02944 \, \text{S} & 0.4706 \end{bmatrix}$.